MA 17: HOW TO SOLVE IT
PROBLEM SET 2

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Due: 11:59PM of October 21. Late submissions will not be accepted.

Instructions:

- *Please submit solutions to at most 3 problems, written carefully. You are encouraged to work on more, but the TA will provide feedback on only 3. I recommend selecting the 3 for which you most want feedback.*
- *Collaboration on homework before you have solved at least 3 problems independently is discouraged, though not strictly forbidden. You should make a serious effort to work through each problem independently before turning to classmates, using the internet, or asking AI.*
- *As long as your submission clearly shows attempts on at least 3 problems, you will earn 8 points toward the final grade. The correctness of your solutions will not affect your points, although feedback will be provided.*
- *Submission should be made through Canvas.*

Problem 1. (Putnam 1979, A4)
Let A be a set of $2n$ points in the plane, no three of which are collinear. Suppose that n of them are colored red and the remaining n blue. Prove or disprove: there are n closed straight line segments, no two with a point in common, such that the endpoints of each segment are points of A having different colors.

Problem 2.
Among positive integers with exactly 28 positive divisors, find the smallest.

Problem 3. (Putnam 2016, A1)
Find the smallest positive integer j such that for every polynomial $p(x)$ with integer coefficients and for every integer k, the integer

$$p^{(j)}(k) = \frac{d^j}{dx^j} p(x)\bigg|_{x=k}$$

(the j-th derivative of $p(x)$ at k) is divisible by 2016.

Problem 4. (Putnam 1969, B1)
Let n be a positive integer such that $n+1$ is divisible by 24. Prove that the sum of all the divisors of n is divisible by 24.

Problem 5. (Putnam 2019, A1)
Determine all possible values of the expression

$$A^3 + B^3 + C^3 - 3ABC$$

where A, B, and C are nonnegative integers.

Problem 6. (Putnam 1983, A3)
Let p be in the set $\{3, 5, 7, 11, \ldots\}$ of odd primes and let

$$F(n) = 1 + 2n + 3n^2 + \cdots + (p-1)n^{p-2}.$$

Prove that if a and b are distinct integers in $\{0, 1, 2, \ldots, p-1\}$ then $F(a)$ and $F(b)$ are not congruent modulo p, that is, $F(a) - F(b)$ is not exactly divisible by p.

Problem 7.
Let p be an odd prime.

(a) Use a primitive root modulo p to prove that $(p-1)! \equiv -1 \pmod{p}$.

(b) Integers $a_1, \ldots, a_p$ form an arithmetic progression whose common difference is not divisible by p. Prove that for some $i \in \{1, \ldots, p\}$,

$$p^2 \mid a_1 a_2 \cdots a_p + a_i.$$

Problem 8. (Putnam 1972, A5)
Show that if n is an integer greater than 1, then n does not divide $2^n - 1$.

Problem 9.
Prove that infinitely many composite positive integers n satisfy $n \mid 2^n - 2$.

Problem 10. (Putnam 2022, A3)
Let p be a prime number greater than 5. Let $f(p)$ denote the number of infinite sequences $a_1, a_2, a_3, \ldots$ such that $a_n \in \{1, 2, \ldots, p-1\}$ and $a_n a_{n+2} \equiv 1 + a_{n+1} \pmod{p}$ for all $n \geq 1$. Prove that $f(p)$ is congruent to 0 or 2 (mod 5).